Drop-test sheet

swift A and B, flat kit F45 F60 B45 | date _____

Goal

Release nose-down at rest from the quad: pulls out unaided, how much height it takes, then a straight, steady glide. Nothing has been flown; all predictions are model values.

Settings

Article	CG range, mm aft of root LE (nominal)	Bend, deg	Steps either side	Mass g	Swoop depth m
swift A foam 400	34.6 to 37.9 (37.0)	6	5 / 7.5	9.8	5 to 18
swift A printed 250	21.6 to 23.7 (23.1)	6	5 / 7.5	9.3	11 to 28
swift B foam 400	40.0 to 43.3 (42.4)	8	7.5 / -	9.4	4 to 17
swift B printed 250	25.0 to 27.1 (26.5)	8	7.5 / -	8.9	11 to 27
kit F45 (250)	36 to 49 (NP 44.0)	6 (tab)	4 / 8	kit table	5 to 22
kit F60 (250)	34 to 44 (NP 39.7)	6 (tab)	4 / 8	kit table	5 to 30
kit B45 (400)	57 to 78 (NP 70.3)	6 (tab)	4 / 8	kit table	2 to 12

Swift: 15% to 8% static-margin CG window, nominal 10% MODEL (NP 30.1% MAC A, 29.1% B); bend is the gauge angle nearest the 10% value (6.0 / 7.9 deg) MODEL; mass CALCULATED (PLA density ASSUMED); depth = formula up to formula x 1.31 (printed) or 2.05 (foam) x 1.95 (50% tail) MODEL. Kit: CG range JUDGEMENT, NP MODEL; 6 deg tab JUDGEMENT; depth over settled, unflagged cases MODEL. Order: forward CG first, step aft 2 to 3 mm JUDGEMENT, then bend one step either side; 2 drops per setting minimum JUDGEMENT.

Bench check

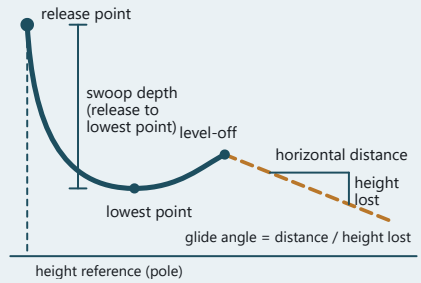
- ☐ Mass: within 5% of table
- ☐ CG: within 1 mm of target
- ☐ Bend: read on gauge
- ☐ Warp / twist: tips level
- ☐ Symmetry: matches template
- ☐ Hook / string: unchanged

Trim guide

Symptom (hand glide or drop)	Adjustment
Dives / no recovery	CG aft one step, or bend up one step
Porpoise / stall / growing oscillation	CG forward one step, or bend down one step
Turns one way	Fix warp first, then a small trailing-edge trim
Wanders or spirals	Tip fins or a small fin
Swoop deeper than predicted	Check bend on the gauge and CG on the balance

Procedure

1. Servo test with propellers off; drop zone clear of people; keep clear of the quad.
2. Steady hover; release only when the article hangs still; abort if it swings.
3. Record the drop height against a fixed reference; film side-on with it in view.
4. One change between drops.
5. Hand-glide low over grass first as a trim check; log it as H.



Model limits

- Force data: plates at Re 60,000 MEASURED; swift outlines are swept and tapered at Re 15,000-40,000 MODEL.
- Pitch-only simulation from rest in still air: no yaw, roll, downwash or string.
- Near-neutral CG unresolved: 423 of 707 cases flagged; 76 of 88 swift cases MODEL.
- Tail-bend effect rests on an inviscid tail in the wake, Re near 10,000; unvalidated MODEL.

Sources: results/runs/rigid_glider/bird_outline_v4_swift, flat_test_v2, pullout; angle gauge: field_pack.pdf p. 4

built 2026-10-14

Drop log

Date		Site		Wind		Height reference	

Drop log

#	Article	CG mm	Bend deg	Height m	Outcome	Swoop depth m	Glide dist m	Clip	Note
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									
12									

G pulls out and glides PS pulls out, then porpoises or stalls D dives, no recovery T tumbles
SP spirals or turns away RF release fault H hand glide (trim check)

Swoop depth: lowest point below the release point (video). Glide dist: horizontal distance over the steady glide.

Best setting found

Article	CG mm	Bend deg	Comment
swift A foam 400			
swift B foam 400			
swift A printed 250			
swift B printed 250			
kit F45			
kit F60			
kit B45			